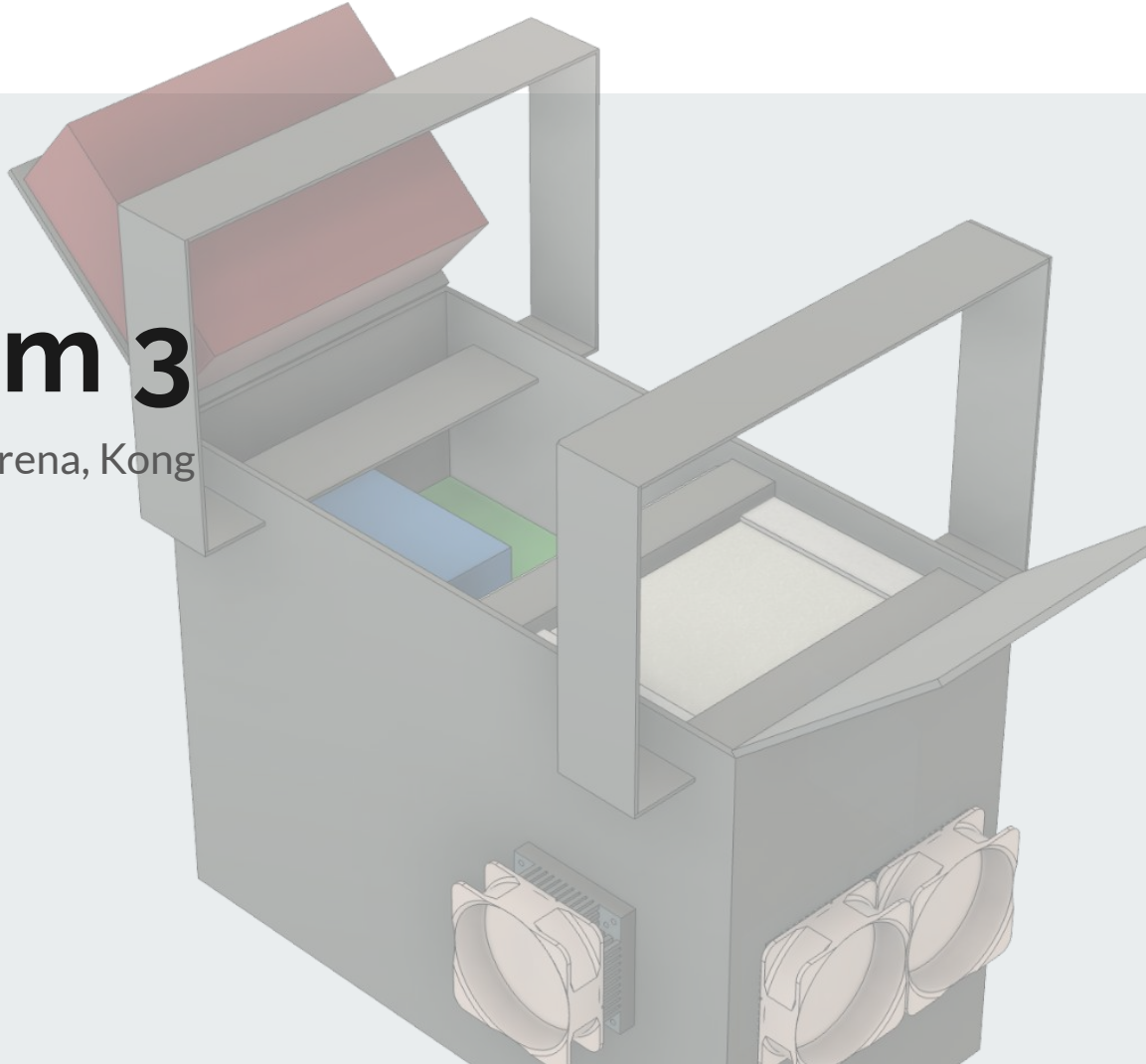




D-PRO: Team 3

Elizabeth, Emma, Enzo, Ryan, Serena, Kong



Our Team



Elizabeth

Program Overseer



Emma

Head of Thermal
Compliance



Enzo

Lead Design Engineer



Ryan

Chief Project Engineer



Serena

Senior Structural Engineer

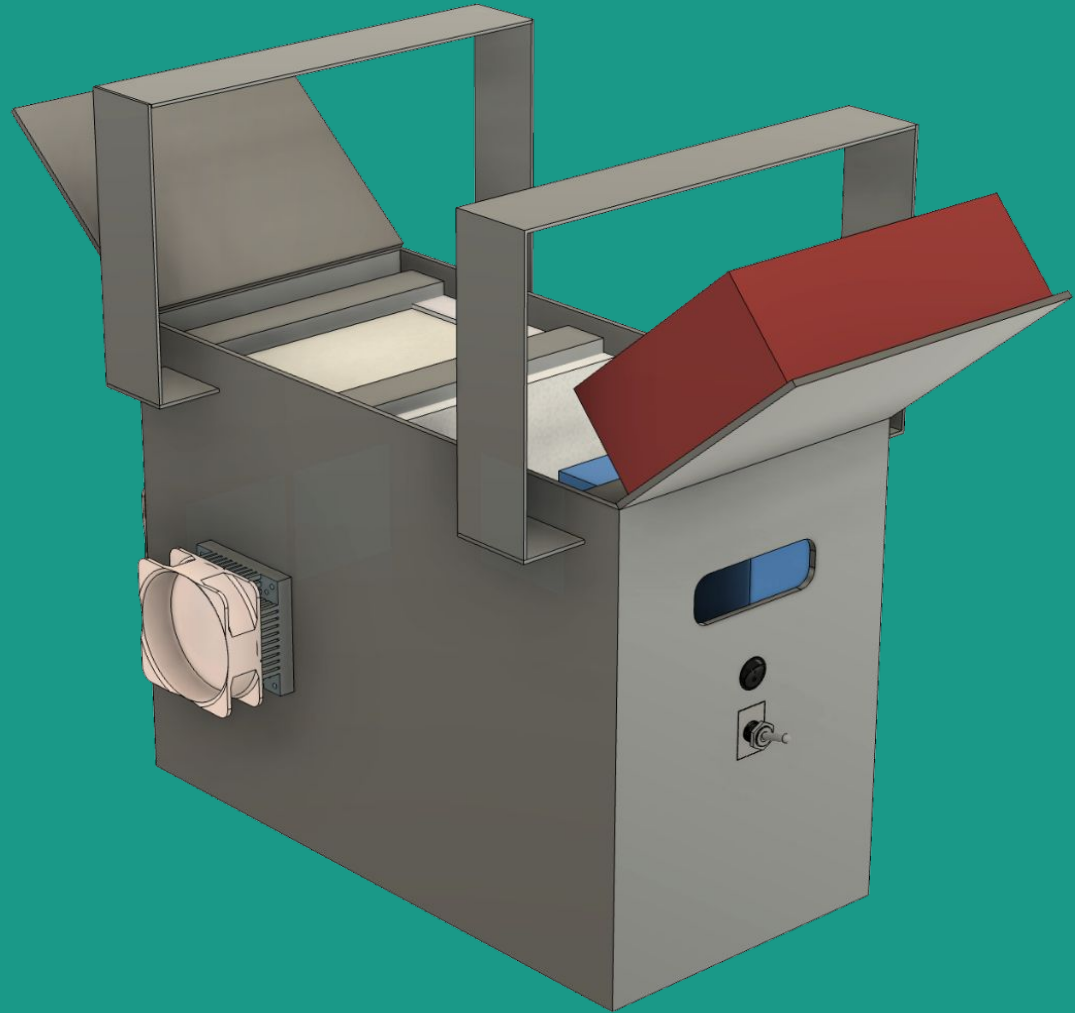


Kong

Chief Product Engineer

Contents

- Function
- Form
- Optimization
- Performance Analysis
- Why Us?



FUNCTION

A compact, insulated organ transport container with active temperature control designed to preserve organ viability during prolonged transport.





Current Function Breakdown

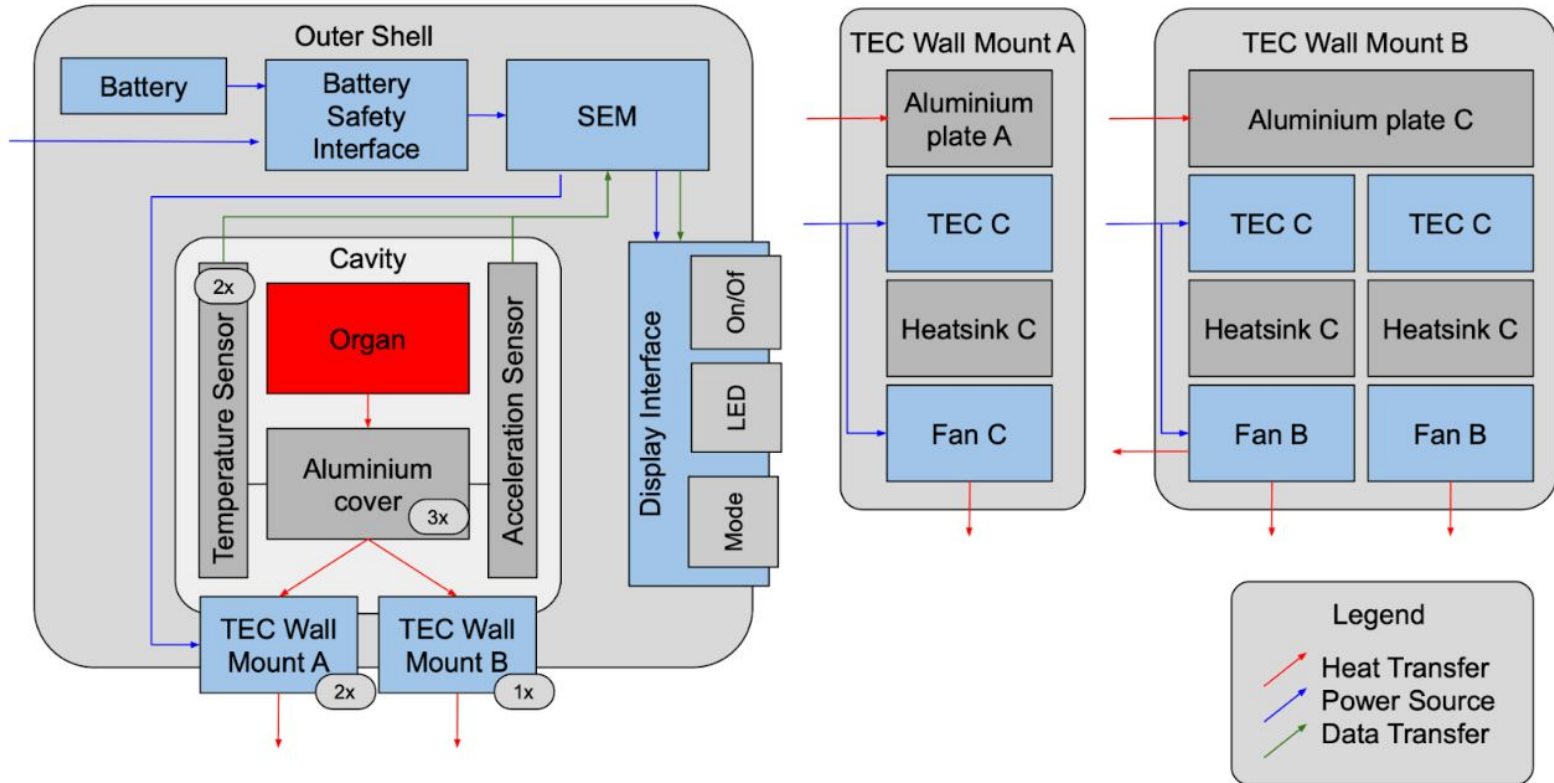
- **5.25 minute cooldown** to 8 degrees celsius
- **4.91g acceleration** for a 20 in drop
- **5.14 hour** battery life
- **Normal and power saver modes** controlled with external switch
- **Colored indicator LEDs** for internal temperatures and drop-shock status
- **Ergonomic handles** for easy carrying
- **14x14x12 cm internal cavity** to accommodate various organ sizes

FORM

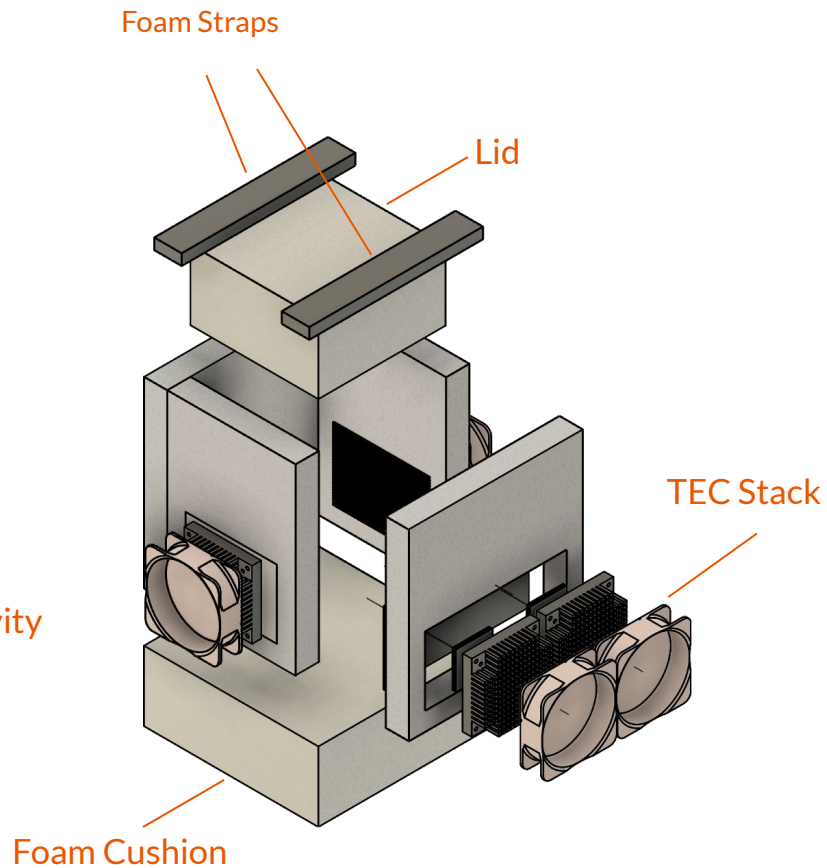
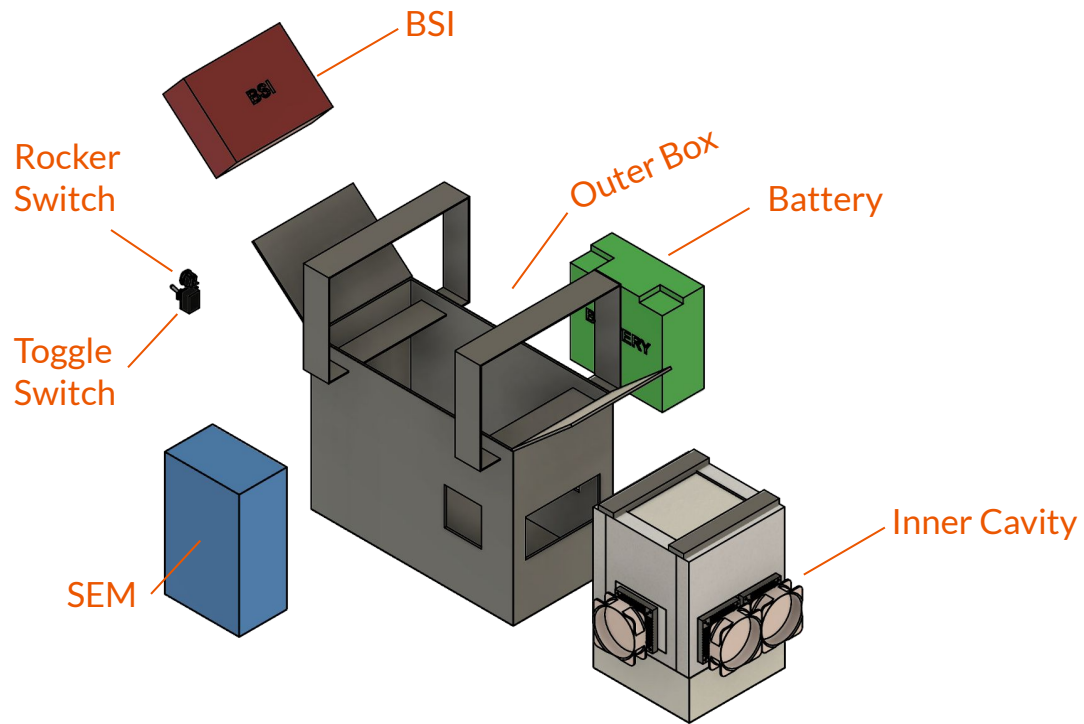
Technical breakdown of our design



Block Diagram

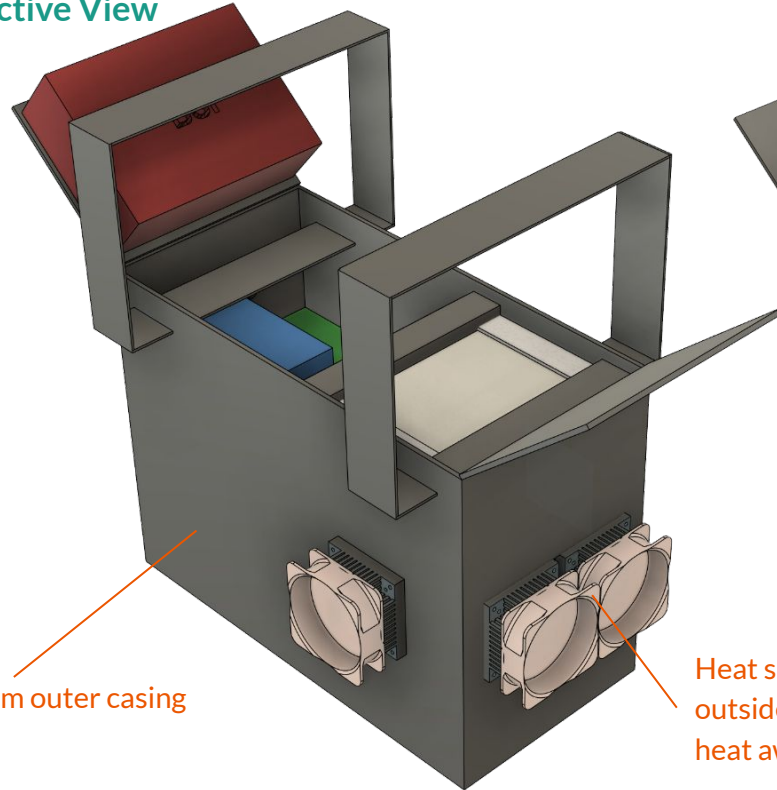


Exploded View



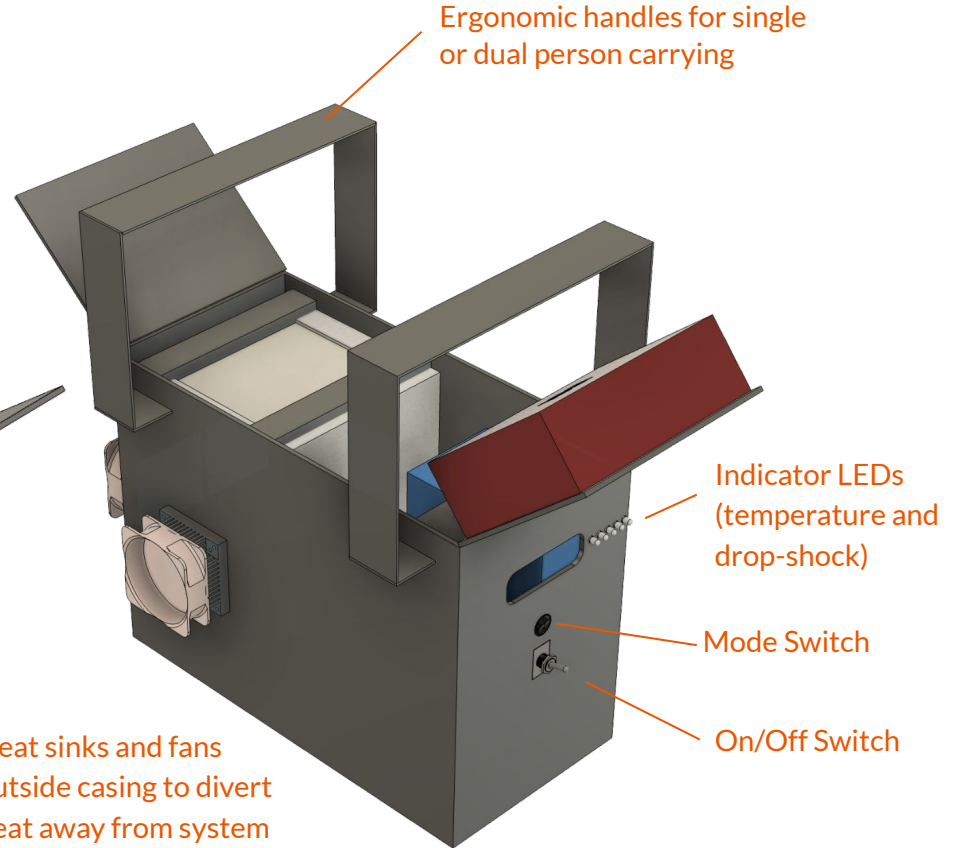
Outer Casing

Perspective View



40 x 30 x 20 cm outer casing

Heat sinks and fans
outside casing to divert
heat away from system



Ergonomic handles for single
or dual person carrying

Indicator LEDs
(temperature and
drop-shock)

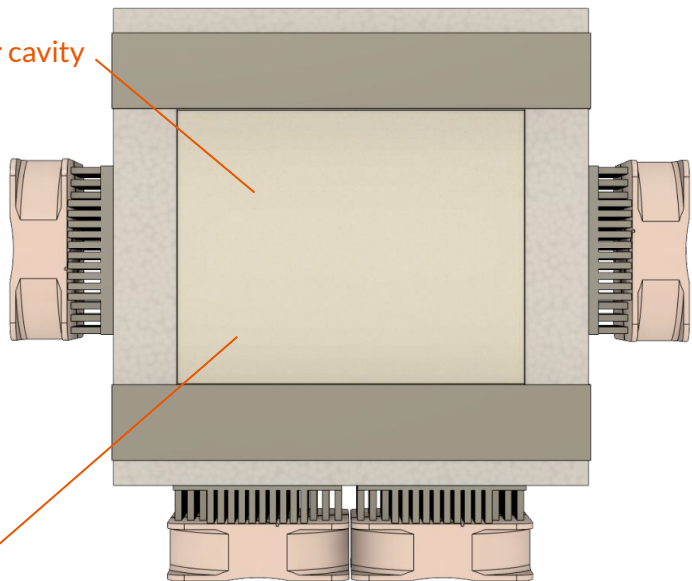
Mode Switch

On/Off Switch

Inner Cavity

Top View

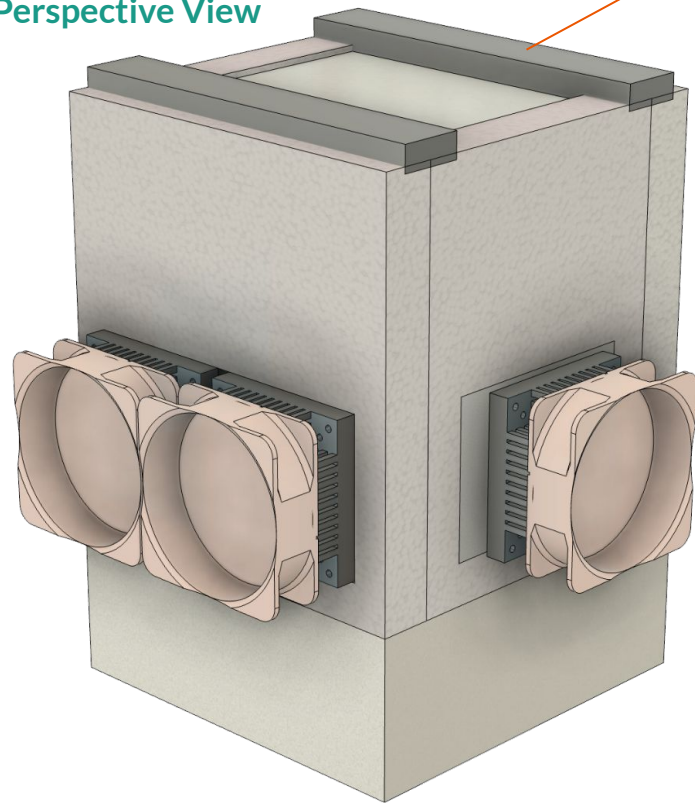
14x14x12 inner cavity



Insulated Cover

Perspective View

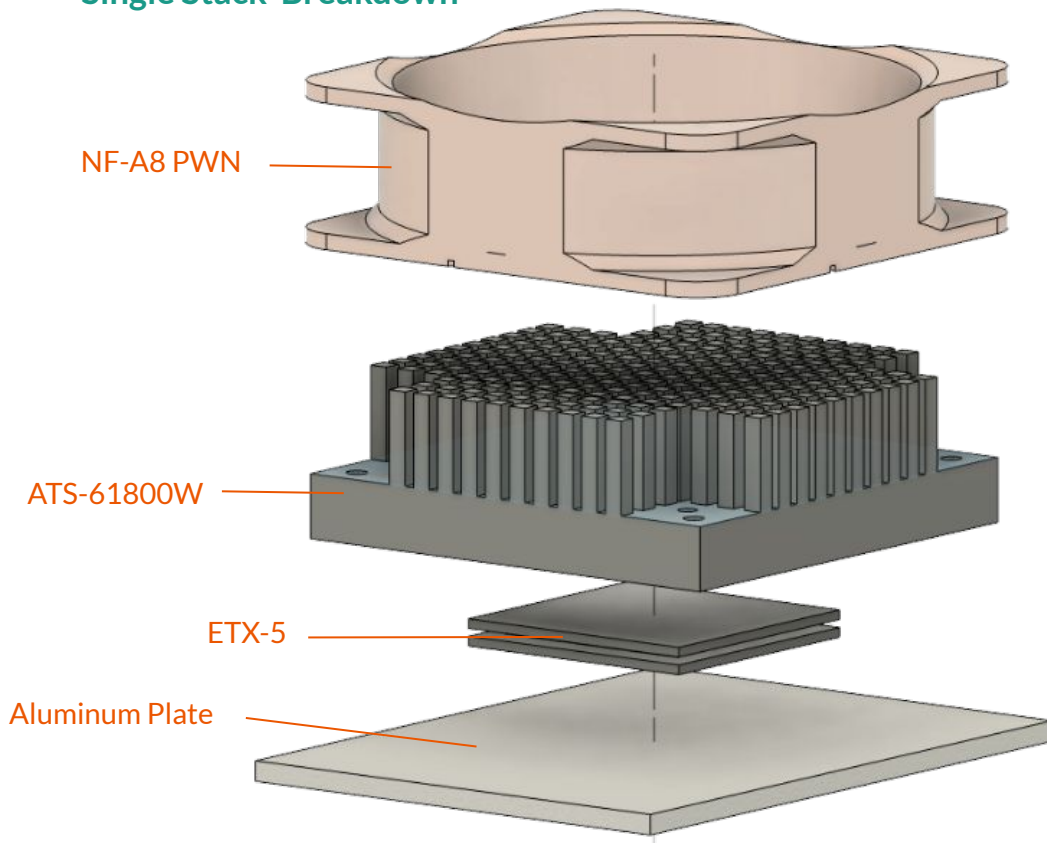
Foam Straps



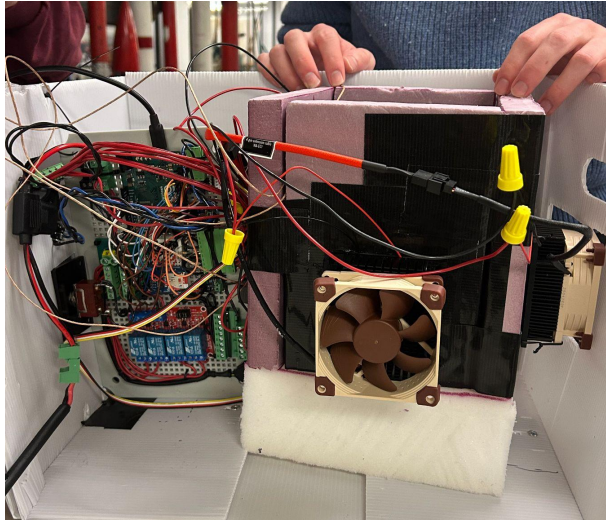
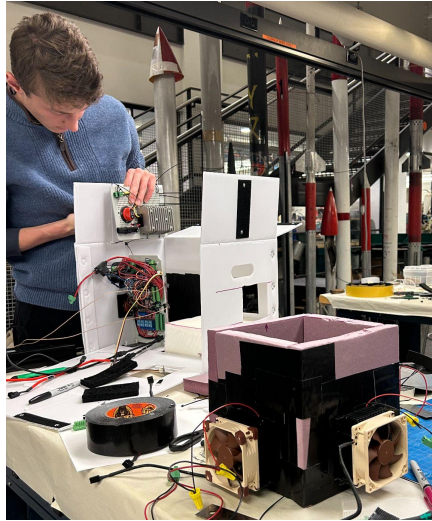
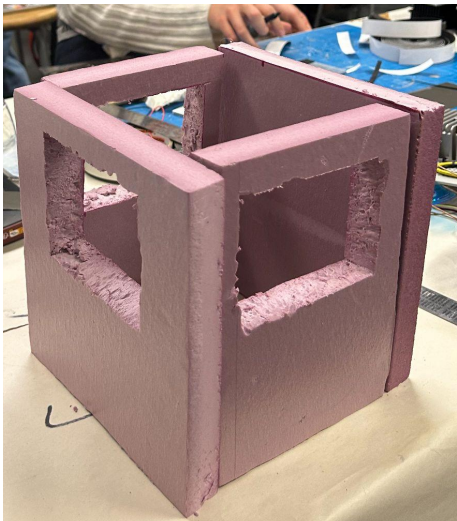
TEC Stacks

- 4 TEC Configuration
- Two Varieties of Stack
- Single Stack (2 TECs on 2 plates):
 - Type C Fan
 - Type C Heatsink
 - Type C TEC
 - Type A Aluminum Plate
- Double Stack (2 TECs on 1 plate):
 - Type B Fan
 - Type C Heatsink
 - Type C TEC
 - Type C Aluminum Plate

Single Stack Breakdown



Prototyping Images

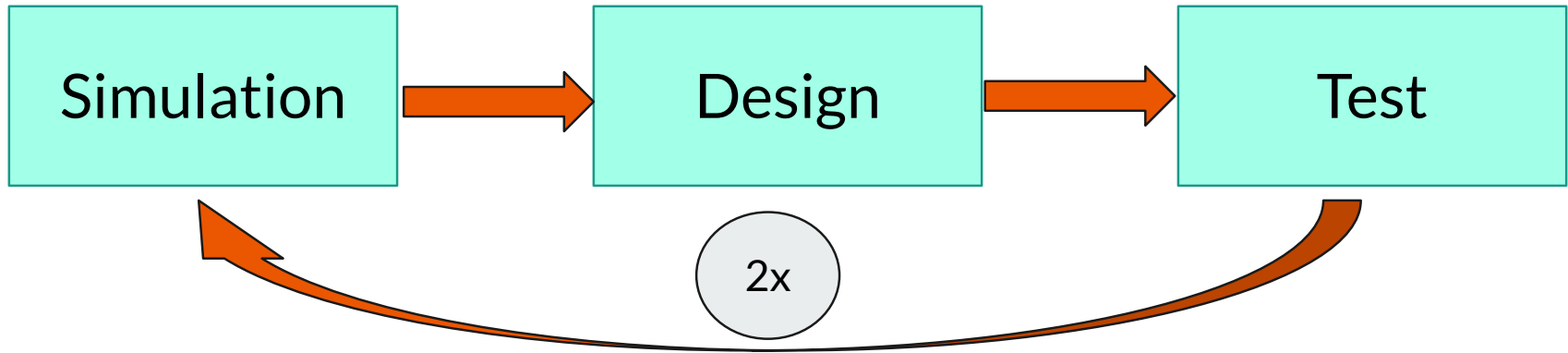


OPTIMIZATION

Overview of simulated analysis and improvements post V1 testing



Design Process





Optimization

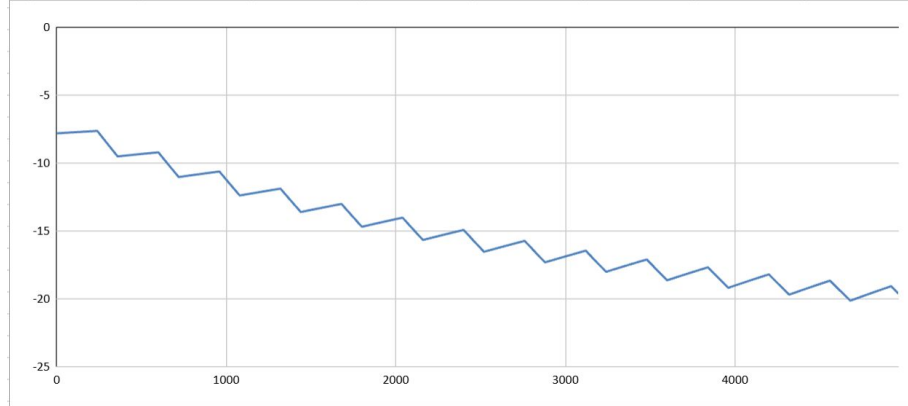
- **V1 tests were unsuccessful**
 - Organ cavity too large to cool
 - Trapped heat
- **Tried a 3 TEC system**
 - Poor TEC adhesion resulting in subpar performance
- **Returned to a 4 TEC system (2 single stacks, 1 dual stack)**
 - Significant improvement in cooldown time
- **Analysis predicted high level performance for V1**
 - Saved expenses on fans, splurged on heatsinks
 - Low temperatures on power saver (no TECs on)
 - High performing battery life
 - Analysis did not perfectly reflect testing due to insulation variability*
- **Less reliance on Analysis for V2**
- **Team-made analysis to optimize cost per cooldown time**
 - TEC type C more efficient over type D
 - Chose a resistance that optimized performance while staying cost effective
- **After testing and not finding success, we added an additional TEC stack ... V3!**



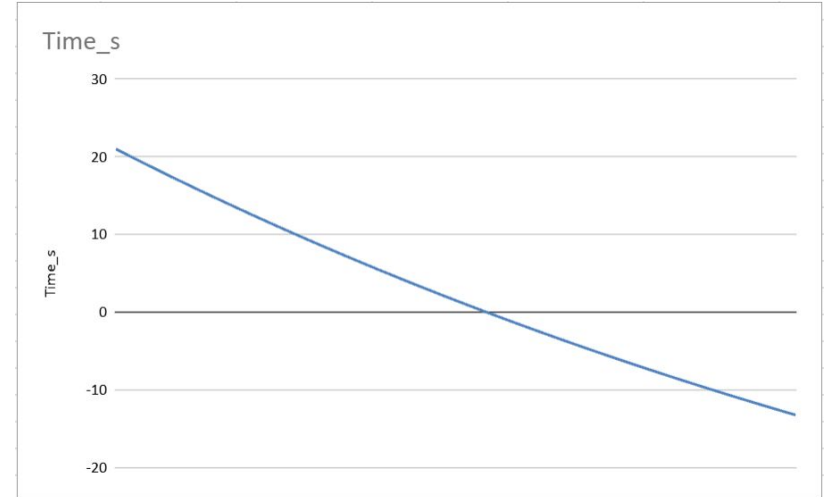
Heat Transfer Analysis

G=MLT-provided thermal analysis program

Power Saver Mode



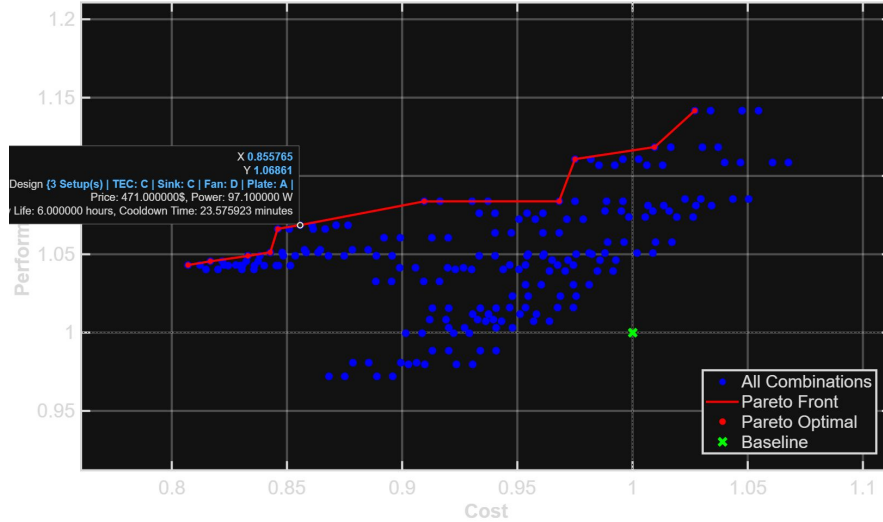
Cooldown Time



Pareto Front Analysis

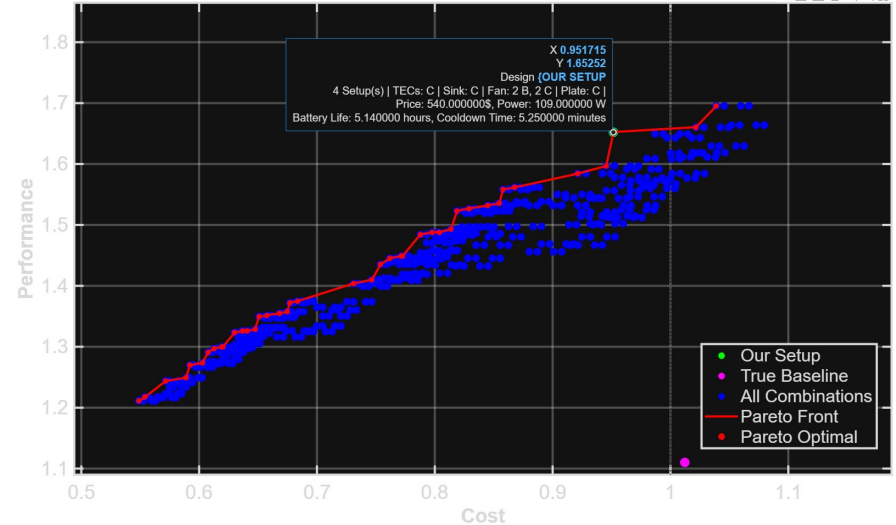
Elizabeth developed a team-specific pareto front analysis to help us better optimize our design choices

Cost vs Performance - All Combinations



V2

Cost vs Performance - All Combinations



V3

PERFORMANCE ANALYSIS

Final Version Testing Results



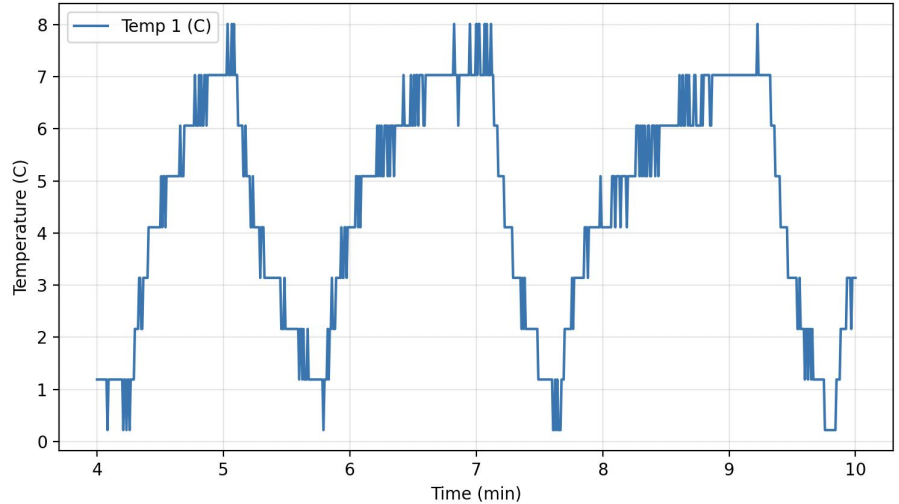


Power Saver

System warmed quickly in power saver mode

Future Improvements:

- Optimize Controls
- Amend Power Saver Configuration (2 TECs on instead of 1)

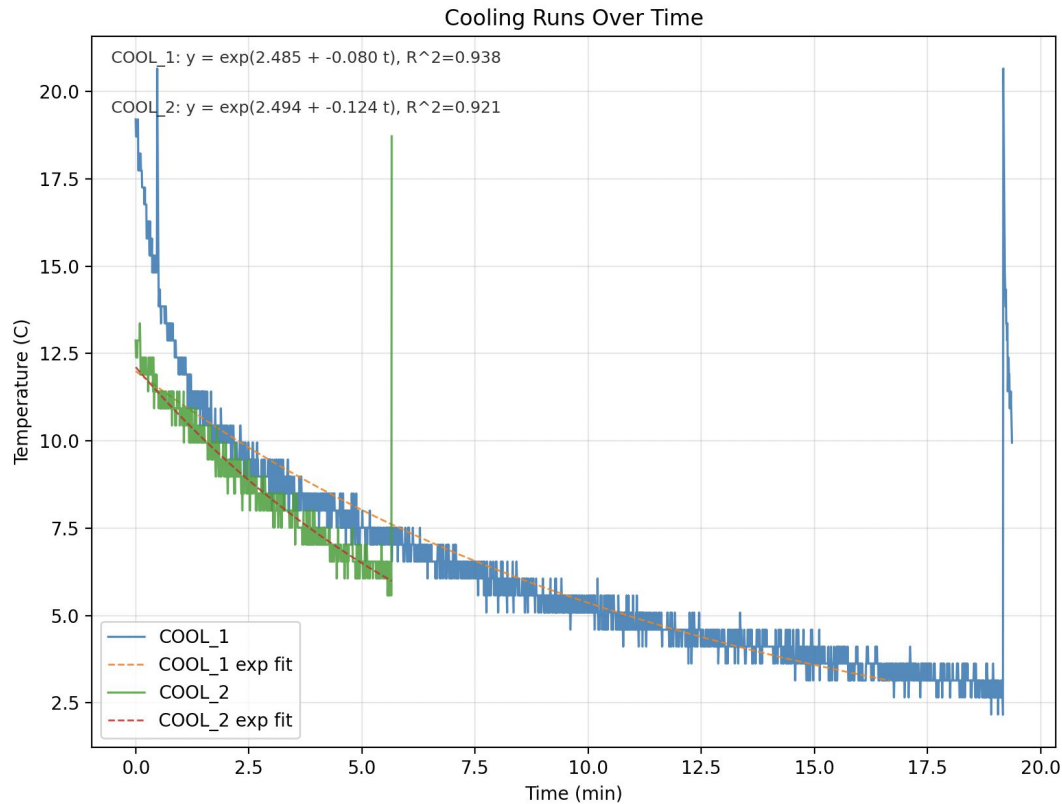




Cool Down Time

5.25 min

Baseline: 17.3 min

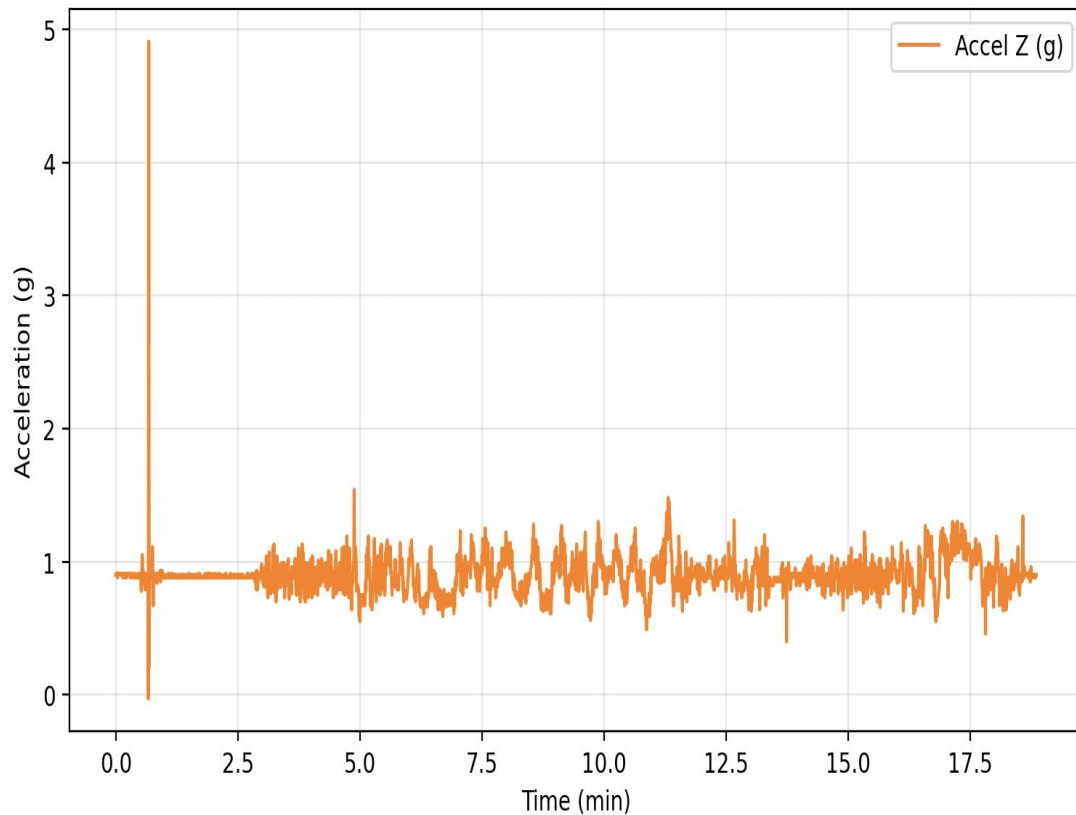




Drop Test

4.91 g

Baseline: 9g

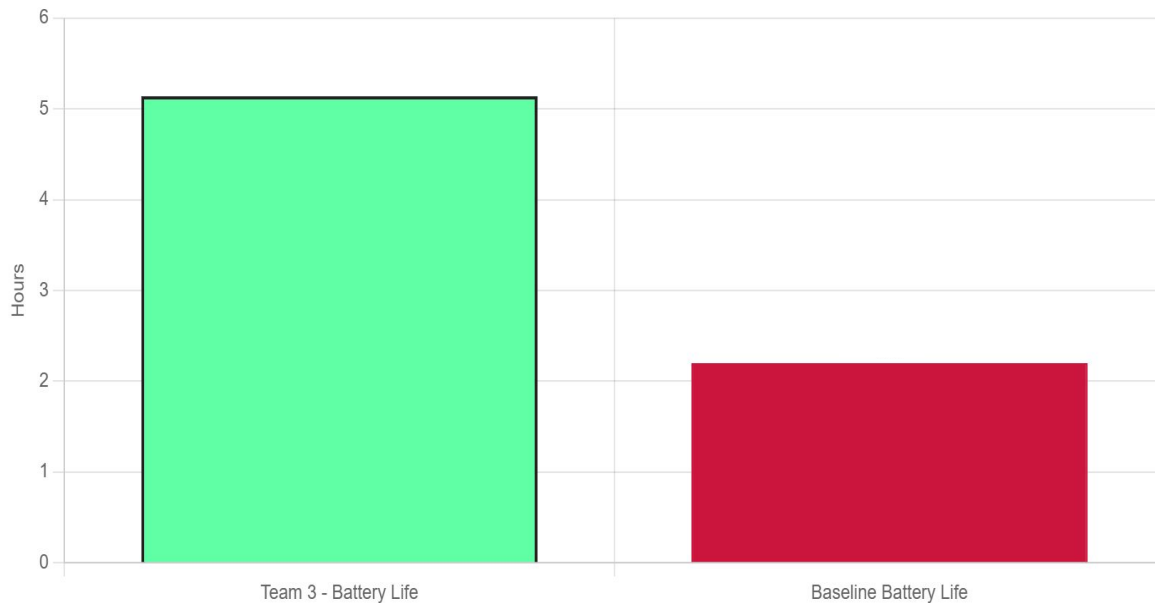




Battery Life

5.14 hours

Baseline: 2.2 hr



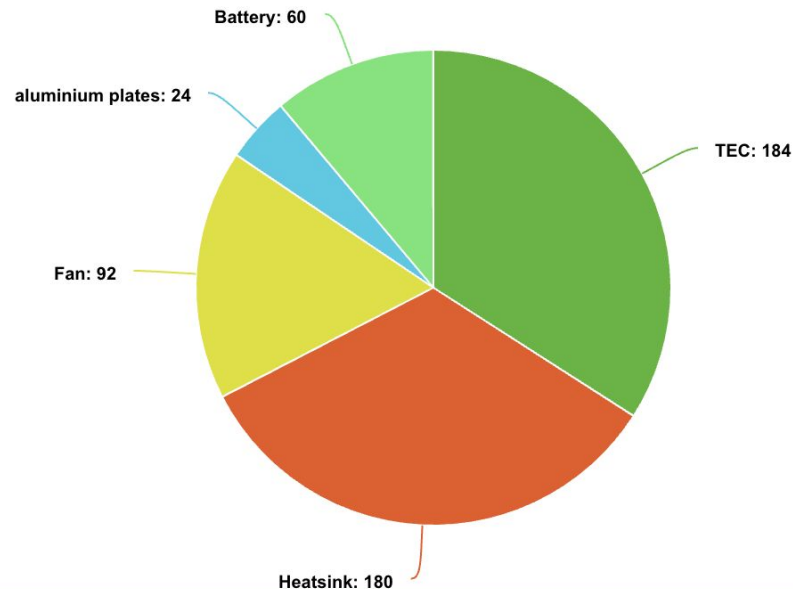
✓ Cost Estimation

\$540

Baseline: \$578

Non-recurring R&D Cost: \$32k

Cost of producing 100,000 units in CY2027: \$35M
(95% learning rate plus fees)



Why Us?



Why Us?

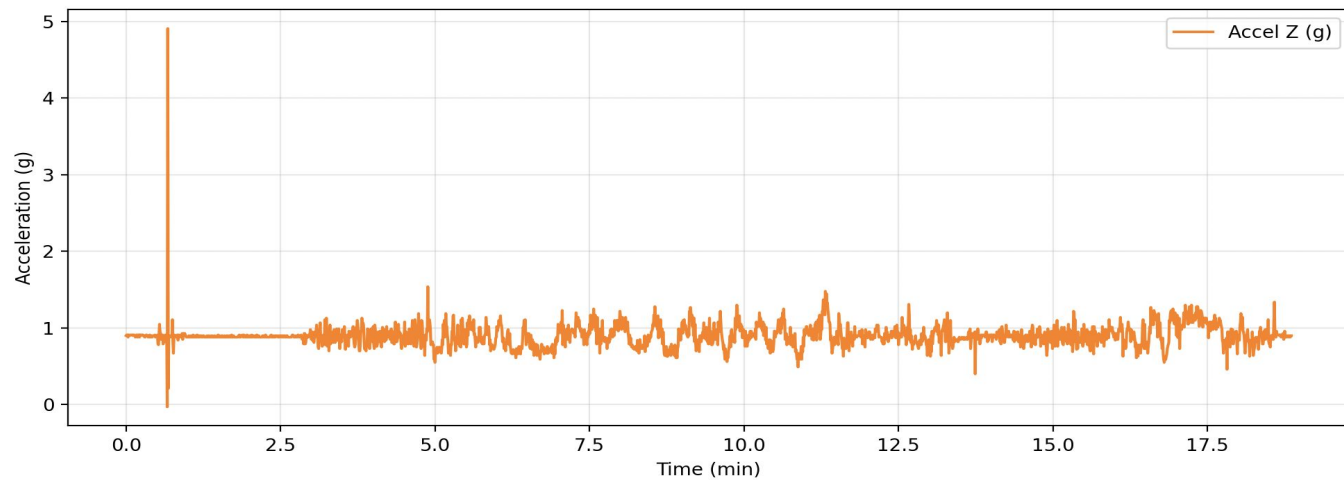
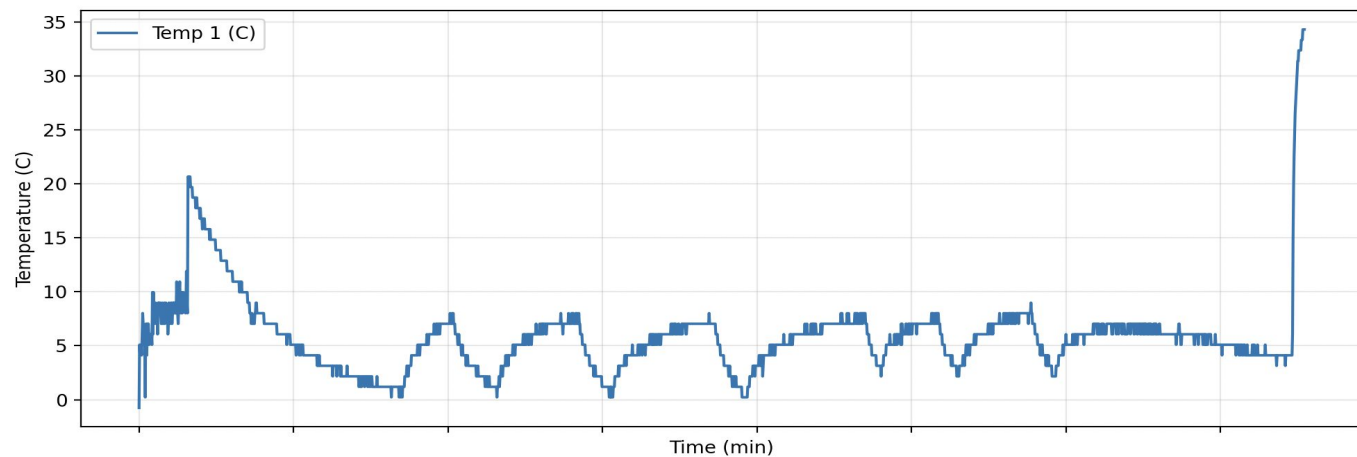
- Delivered a system that exceeds baseline in all performance objective metrics.
- Displayed capacity to design, engineer, and validate a to-spec system.
- Applied our principled method for overcoming engineering challenge

Questions?

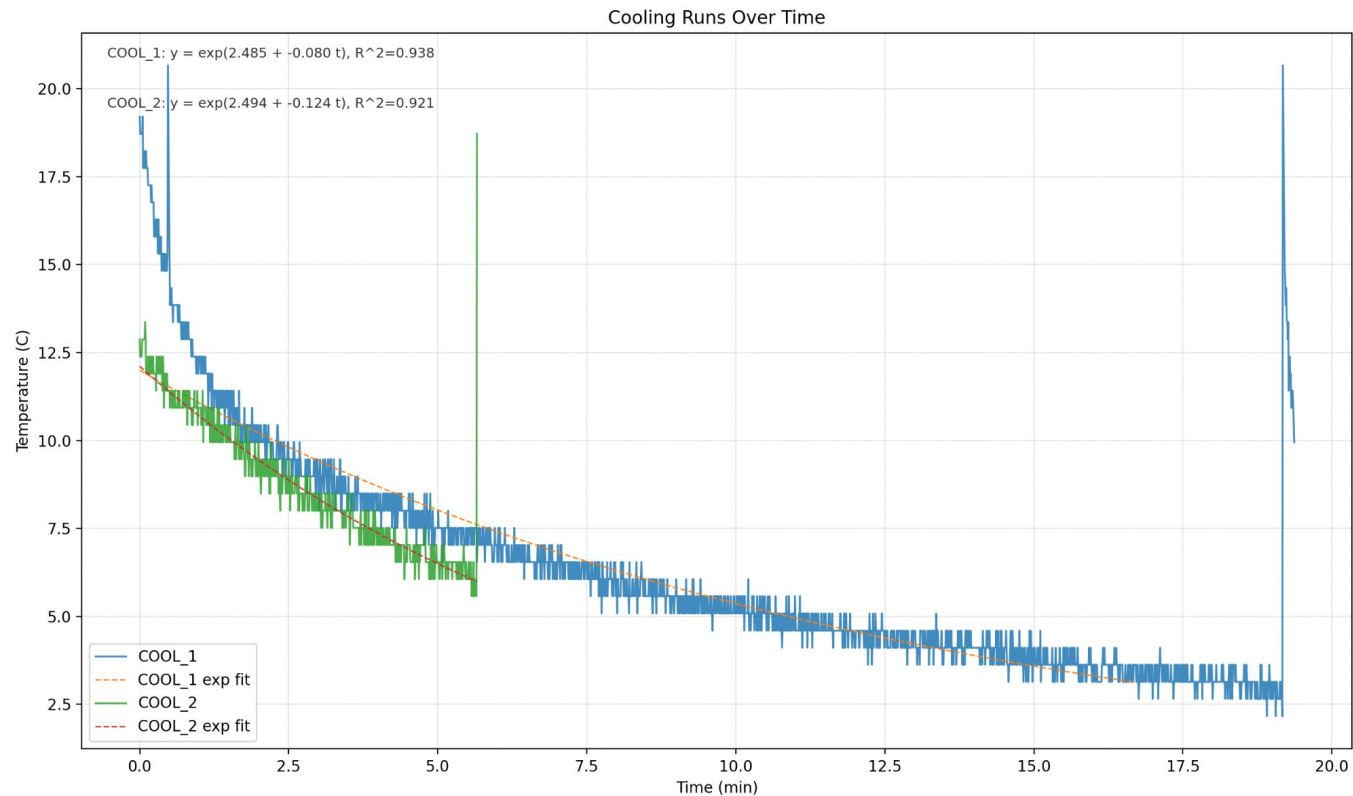
Appendix



Journey



Cool Down Data



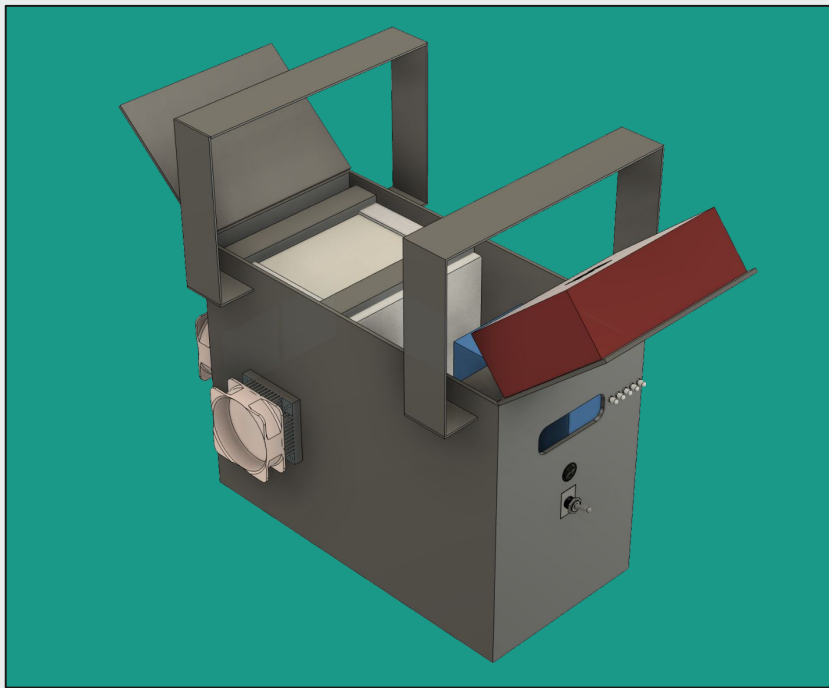
Baseline Performance

1. Power Draw: 122.4 W
2. Weight: 6.9 kg
3. Cooldown Time: 17.3 mins
4. Drop Test: 9 G
5. **Battery Life: 133.1 mins— just over 2 hours!**

Our Performance

1. Power Draw: 109 W
2. Weight: 7.23 kg
3. Cooldown Time: 5.25 mins
4. Drop Test: 4.91 G
5. Battery Life: 5.14 hours

How it Performs Against the Requirements?



Successfully keeps it in the 4 to 8 degree range under normal circumstances via the use of 4 Type C TECs, 4 Type C Heatsinks, 2 Type B Fans, and 2 Type C Fans

Cools down in 5.25 minutes

Is well within the weight and size requirements

Perfectly sealed

Contains all necessary sensors

Vertical Drop Shock Test resulted at only 4.9g

Battery lasts for 5.14 hours